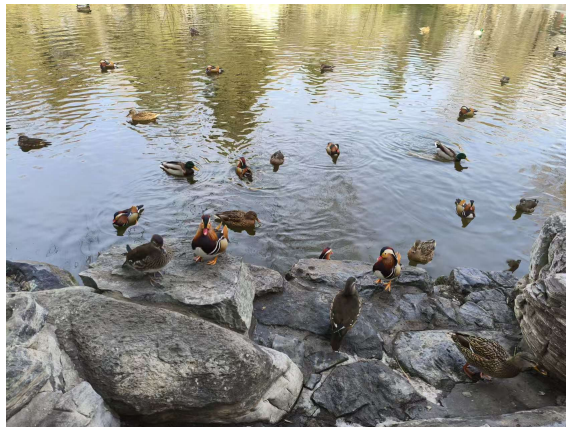


YUANYANG.KEM

Key Encapsulation Mechanism

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Abstract. This document specifies YUANYANG.KEM, a lattice-based key encapsulation mechanism following the NTRU encryption paradigm. YUANYANG.KEM offers competitive performance. Its public key and ciphertext sizes are comparable to, or smaller than, those of Kyber, while key generation, encapsulation and decapsulation are simple and fast. At the same time, YUANYANG.KEM attains its target security levels under the conservative Core-SVP model and drives the estimated decryption-failure probability below the corresponding target bound, providing strong end-to-end security guarantees.

As its name indicates, YUANYANG.KEM is the KEM companion of the lattice-based signature scheme YUANYANG.DSA: a single YUANYANG.DSA key pair can be used unchanged for encapsulation and decapsulation, without altering algorithms or parameters. This shared-key feature makes the pair YUANYANG.KEM/YUANYANG.DSA particularly attractive for real-world applications such as payment cards and end-to-end-encrypted messaging, and constrained environments where key management is at a premium.

Keywords: Key encapsulation mechanisms · Lattice-based cryptography · NTRU lattices

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1 Introduction

Lattice-based KEMs offer one of the most attractive trade-offs between size and speed among post-quantum candidates: they are typically faster than RSA and ECC while keeping public key and ciphertext sizes broadly comparable to those of RSA. Concrete instantiations build on LWE, NTRU, or their structured variants, and the same families of hard problems also yield practical signatures.

In current deployments, however, the post-quantum KEM and the signature live entirely separate lives. The two NIST standards CRYSTALS-Kyber [SAB⁺20] and CRYSTALS-Dilithium [LDK⁺20] use different parameters even at matching security levels, so their key pairs cannot be reused across primitives. This contrasts sharply with the RSA and ECC world, where a single key pair routinely supports both encryption and signing. Recovering this property in the post-quantum setting is more than an aesthetic concern: it directly affects scenarios such as payment cards [DLP⁺12] and secure group communication [TSPP21, FGPZ25], and the resulting savings in key distribution and storage are especially valuable in constrained environments.

Motivated by this gap, we design YUANYANG, a suite of NTRU-based KEM and signature schemes built so that a single key pair serves both. The signature YUANYANG.DSA relies on a high-quality NTRU trapdoor; the KEM YUANYANG.KEM only needs a single short vector to decrypt, exactly as in classical NTRU encryption. As a consequence, the two algorithms can share keys without compromise: YUANYANG.KEM reuses any YUANYANG.DSA key pair as is, and when used standalone its key generation simplifies considerably.

Beyond key reuse, YUANYANG.KEM is competitive as a standalone lattice-based KEM. It offers better compactness than Kyber, together with simple and fast operations. In addition, YUANYANG.KEM is engineered for conservative security. In contrast with current NTRU-based KEM schemes [CDH⁺20, LS19, ZFY23, LZL⁺25], YUANYANG.KEM adopts dense secret and randomness coefficients to defeat small-secret attacks [How07, ETWY22, KKNM25]. To this end, it uses the small polynomial $p = 1 + X^{d/2}$, of coefficient norm $\sqrt{2}$, as the NTRU masking modulus and adds a lightweight overflow-correction step in decryption, reducing the decryption-failure probability. With these design choices, the three parameter sets target 128, 256, and 512 bits of classical security under the Core-SVP model, with decryption-failure probabilities below the corresponding target bounds.

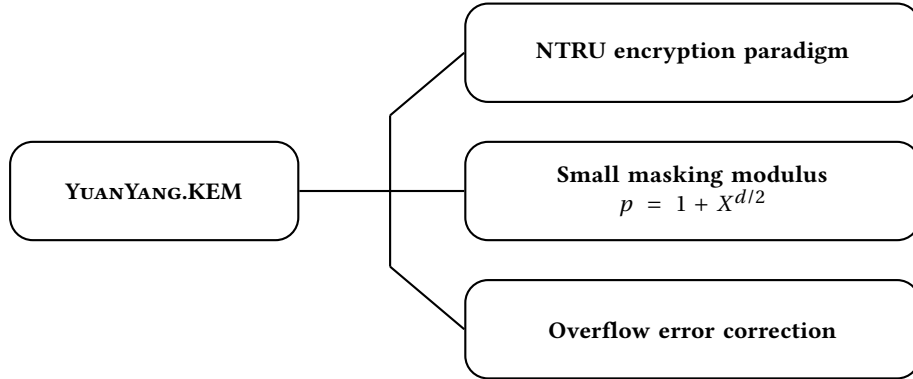


Fig. 1: Main components of YUANYANG.KEM.

2 Preliminaries

Vectors are denoted by bold lower-case letters and are always considered as column vectors; matrices use bold upper case. We say that a matrix is a basis of a space when its columns form a basis. For a vector $\mathbf{x} = (x_1, \dots, x_d)$, $\|\mathbf{x}\| = (\sum_i |x_i|^2)^{1/2}$ is its Euclidean norm and $\|\mathbf{x}\|_\infty = \max_i |x_i|$ its infinity norm.

Lattices. A lattice is a discrete subgroup of $\mathbb{R}^n$ or, equivalently, the set of *integer* linear combinations of a basis $\mathbf{B}$ of $\mathbb{R}^n$. Its volume is $\det \mathbf{B}$ and is independent of the chosen basis.

Power-of-two cyclotomic rings. For $d = 2^n$, write $K = \mathbb{Q}[X]/(X^d + 1)$ for the $2d$ -th cyclotomic field. We work mostly in the integer subring $\mathcal{R} = \mathbb{Z}[X]/(X^d + 1)$, and occasionally in the real overring $K_{\mathbb{R}} = \mathbb{R}[X]/(X^d + 1)$. Let $\zeta_j = \exp(i(2j-1)\pi/d)$ for $1 \leq j \leq d$ denote the roots of $X^d + 1$, equivalently the primitive $2d$ -th roots of unity, and set $\varphi_j(f) = f(\zeta_j)$ for $f \in K_{\mathbb{R}}$.

Elements of $K_{\mathbb{R}}$ admit several useful representations:

- the *coefficient embedding* $f = \sum_i f_i X^i \mapsto \mathbf{f} = (f_0, \dots, f_{d-1})$;
- the *canonical embedding* $f \mapsto \varphi(f) = (\varphi_1(f), \dots, \varphi_d(f))$, also known as the Discrete Fourier Transform (DFT), which sends polynomial multiplication in $K_{\mathbb{R}}$ to coordinate-wise multiplication;
- the *multiplication matrix*

$$f \mapsto [f] := \begin{bmatrix} f_0 & -f_{d-1} & \dots & -f_1 \\ f_1 & f_0 & \dots & -f_2 \\ \vdots & & \ddots & \vdots \\ f_{d-1} & f_{d-2} & \dots & f_0 \end{bmatrix}.$$

When needed, $K_{\mathbb{R}}^{++}$ denotes the cone of self-adjoint elements f whose canonical coordinates $\varphi_i(f)$ are positive real numbers. We have $\|\varphi(f)\| = \sqrt{d} \|\mathbf{f}\|$. The complex conjugate of f is

$$f^* = f_0 - f_{d-1}X - \dots - f_1X^{d-1}, \quad \varphi(f^*) = \left(\overline{\varphi_i(f)}\right)_i,$$

and one checks that $[f^*] = [f]^t$.

NTRU lattices. Let q be a positive integer and let $f \in \mathcal{R}$ be invertible modulo q (equivalently, $\det[f]$ is coprime to q). For some short $g \in \mathcal{R}$, set $h = g/f \bmod q$ and define the NTRU module associated to h ,

$$\mathcal{M}_{\text{NTRU}} = \{(u, v) \in \mathcal{R}^2 : hu - v \equiv 0 \pmod{q}\},$$

together with its lattice avatar

$$\mathcal{L}_{\text{NTRU}} = \{(\mathbf{u}, \mathbf{v}) \in \mathbb{Z}^{2d} : [h]\mathbf{u} - \mathbf{v} \equiv 0 \pmod{q}\}.$$

This lattice has volume q^d . Over $\mathcal{R}$ it is generated by (f, g) together with any pair (F, G) satisfying $fG - gF = q$, so $\mathcal{L}_{\text{NTRU}}$ admits the coefficient basis

$$\mathbf{B}_{f,g} = \begin{bmatrix} [f] & [F] \\ [g] & [G] \end{bmatrix}.$$

A direct computation gives $([h], -\text{Id}_d) \cdot \mathbf{B}_{f,g} \equiv 0 \pmod{q}$, so h plays the role of the public verification element.

The NTRU problem comes in two standard flavors. The *search* version asks, given $h = g/f \bmod q$, to recover any rotated pair $(X^i f, X^i g)$. The *decisional* version asks to distinguish such an h from a uniformly random element of $\mathcal{R}_q := (\mathbb{Z}/q\mathbb{Z})[X]/(X^d + 1)$. Both are conjectured intractable for d large enough, and their hardness underpins the security of YUANYANG.KEM.

3 Design Rationale

YUANYANG.KEM is designed as the KEM paired with the YUANYANG.DSA signature scheme: the two schemes share the same parameters and support secure key reuse. This simplifies joint deployment and opens the door to efficient, versatile protocols in settings such as payment cards [DLP⁺12] and secure group communication [TSPP21, FGPZ25]. Nevertheless, when employed independently as a KEM, YUANYANG.KEM offers **competitive performance** and **strong security** — and this has remained our leading design principle throughout.

To attain high efficiency both in terms of bandwidth and speed, YUANYANG.DSA instantiates the GPV hash-and-sign scheme [GPV08] over NTRU lattices. This gives two options for the design paradigm of the companion KEM: classical NTRU encryption [HPS98] and BAT-style encryption [FKPY22]. The latter uses the full NTRU basis to decrypt, which is able to recover larger encryption errors and thus yields smaller size and higher security. On the downside, its key generation that requires to compute a full NTRU basis is expensive to a standalone KEM especially when perfect forward secrecy is taken into account. YUANYANG.KEM therefore follows the classical NTRU paradigm, trading some bandwidth for simpler and faster standalone key generation.

3.1 Classical NTRU Encryption

The first NTRU encryption scheme was invented by Hoffstein, Pipher and Silverman in 1996 [HPS98]. Since then, a number of variants [SS11, BBC⁺20, DHK⁺23, ZFY23, LZL⁺25] have been proposed for improved security and efficiency. For a typical NTRU encryption scheme, the secret key is a pair of short polynomials (f, g) in the ring $\mathcal{R} = \mathbb{Z}[X]/(\phi(X))$, and the public key is $h = g/f \bmod q$. The masking modulus p is a small polynomial, for example $p = 2$ or $p = 3$, and the message space is $\mathcal{R}/p\mathcal{R}$.

The basic idea of the encryption procedure is to mask the message m (scaled by p) with a pseudo-random polynomial $hs + e$ with small (s, e) . Sometimes, the term e is omitted and m assumes the same role. The ciphertext is

$$c = hs + e + m/p \bmod q.$$

The decryption first computes an intermediate polynomial $\bar{c} = (pfc \bmod q)$, then

$$\bar{c} = p(gs + fe) + fm \bmod q.$$

Correct decryption requires all coefficients of $p(gs + fe) + fm$ to lie in $(-q/2, q/2)$, so that $\bar{c} = p(gs + fe) + fm$ after centered lifting. It then suffices to remove $p(gs + fe)$ by reducing modulo p and to recover m by multiplying by the inverse of f modulo p .

3.2 Designs of YUANYANG.KEM

YUANYANG.KEM uses the power-of-two cyclotomic ring $\mathcal{R} = \mathbb{Z}[X]/(X^d + 1)$ and a modulus q supporting partial NTTs over $\mathcal{R}_q$.

For fixed ring $\mathcal{R}$ and the modulus q , larger (f, g) and (s, e) provide higher security against key recovery and message recovery respectively, but are more likely to cause decryption failures. Achieving the target security level while maintaining a sufficiently low decryption-failure probability is the central design problem for NTRU encryption. To address it, YUANYANG.KEM uses two main techniques.

1. Small masking modulus. YUANYANG.KEM chooses a small polynomial $p = 1 + X^{d/2}$ as the masking modulus, inspired by the NEV algorithm [ZFY23]. In the working ring $\mathcal{R} = \mathbb{Z}[X]/(X^d + 1)$ the monomial $X^{d/2}$ satisfies $(X^{d/2})^2 = X^d = -1$, so it behaves exactly like the imaginary unit, and $p = 1 + X^{d/2}$ is the ring-theoretic counterpart of the Gaussian integer $1 + i$. The familiar identity $(1 + i)(1 - i) = 2$ in $\mathbb{Z}[i]$ then translates into $\mathcal{R}$ as

$$(1 + X^{d/2})(1 - X^{d/2}) = 1 - X^d = 2,$$

so p is a genuine factor of 2, with coefficient norm $\sqrt{2}$. Using p rather than the rational integer 2 as the masking modulus is therefore akin to working with a “half” of 2: the quotient by (p) has size $2^{d/2}$, halfway between the trivial quotient and the quotient by (2) on a logarithmic scale. This buys back a corresponding margin on the ciphertext modulus and keeps the encoding tighter than in classical NTRU, without leaving the cyclotomic ring.

2. Overflow error correction. Error correction has been used to lower decryption-failure rates in many lattice-based KEMs, most notably LWE-based schemes [SCH⁺19, GZB⁺19]. Some NTRU-based schemes [LFZZ24] also use error correction, but impose a strong constraint on f . YUANYANG.KEM instead uses an overflow-correction routine that recovers from at most one coefficient overflow in $p(gs + fe) + fm$. The core of the routine is a fast search technique that is folklore in algebraic lattice sieving but has not previously been used in a lattice-based KEM. Correcting one overflow is roughly equivalent to shrinking the effective error size by a factor of $\sqrt{2}$, which is enough to push the estimated decryption-failure probability below the target bound.

3.3 Criterion for Parameter Selection

YUANYANG.KEM adopts a conservative criterion for parameter selection.

1. Solid theoretical security. YUANYANG.KEM admits a formal security proof based on the well-studied NTRU and Ring-LWE assumptions. At every security level λ , the estimated decryption-failure probability is pushed below $2^{-\lambda}$, which removes observed failures as a practical attack avenue and supplies the correctness parameter used in the FO analysis.

2. Conservative security estimate. The concrete security of YUANYANG.KEM is estimated from the Core-SVP hardness of the best known attacks. Under this conservative model, the three parameter sets of YUANYANG.KEM target 128-, 256-, and 512-bit classical security, respectively; the corresponding quantum Core-SVP estimates are reported in Table 3. In addition, all secret-key and encryption-randomness coefficients have standard deviation at least 1, neutralizing attacks that exploit secret sparsity [How07, ETWY22, KKNM25].

3.4 Decryption Failures

Like most practical lattice-based KEM schemes, YUANYANG.KEM is subject to a very small probability of decryption failures. There is currently no universal criterion for the decryption-failure rate required at a given security level λ : Kyber’s decryption-failure estimates are above $2^{-\lambda}$ for $\lambda = 192$ and 256, whereas HQC gives rigorous bounds below $2^{-\lambda}$. However, several failure-based attacks [DRV20, BS20, DB22] accelerate the search for decryption failures and enable theoretical attacks against lattice-based KEMs. To avoid this risk, YUANYANG.KEM selects concrete parameters whose estimated decryption-failure probability is below the target security bound. We provide a detailed decryption failure analysis in Section 5.5.

3.5 Differences with Comparable Schemes

YUANYANG.KEM follows the 30-year-old classical NTRU paradigm, and some algorithmic details are inspired by recent advances [DHK⁺23, ZFY23, LZL⁺25]. Nevertheless, YUANYANG.KEM has its own distinctive features.

The defining feature of YUANYANG.KEM is that it can be used together with YUANYANG.DSA signatures under the same key pair. This makes YUANYANG of special interest to certain real-world applications. To the best of our knowledge, YUANYANG is the first practical paired post-quantum KEM and signature cryptosystem of this kind. In addition, the use of dense parameters and the lightweight overflow error correction also distinguish YUANYANG.KEM from other NTRU-based KEM schemes.

3.6 Joint Security with YUANYANG.DSA

The main motivation for using YUANYANG.KEM together with YUANYANG.DSA is to reduce the cost of joint deployment. In many applications, a device needs both key establishment and authentication. Using two unrelated post-quantum key pairs increases the public-key distribution, certificate size and secret-key storage. This is especially relevant to constrained platforms, payment cards and secure group communication, where key management can be as important as the cost of encryption and signing.

However, reusing one key pair for two primitives is not automatically secure. A signing oracle and a decapsulation oracle give the adversary two different views of the same secret material. Thus, the security of YUANYANG.KEM and YUANYANG.DSA as standalone schemes does not by itself imply security in the key reuse setting. For this reason, YUANYANG treats the same-key use of the two schemes as an explicit security property, rather than as an implementation shortcut.

The compatibility comes from the common NTRU key structure. Both schemes use the same public key $h = g/f \bmod q$ and the same short secret pair (f, g) . The joint-security statement is given in Section 5.6, where YUANYANG.KEM is analysed with access to a signing oracle for YUANYANG.DSA, and YUANYANG.DSA is analysed with access to a decapsulation oracle for YUANYANG.KEM.

4 Algorithm Description

YUANYANG.KEM is built in two stages. First, an NTRU-style public-key encryption scheme provides the encryption primitives; this is then upgraded into an IND-CCA-secure KEM by a standard CPA-to-CCA transform. We describe the parameters, then the underlying PKE, and finally the KEM construction.

4.1 Parameters

A YUANYANG.KEM instance is fully determined by the following data:

- the working ring $\mathcal{R} = \mathbb{Z}[X]/(X^d + 1)$ with $d = 2^\ell$, the modulus q , and the masking modulus $p = X^{d/2} + 1$;

- the standard deviation σ_f of the secret-key coefficients;
- the standard deviation σ_s of the encryption randomness s and the standard deviation σ_e of the encryption randomness e ;
- the positive rounding parameter k , which forces ciphertext coefficients to lie in $\mathbb{Z}_b$ with $b = \lceil q/k \rceil$.

Concrete values for the three target security levels are given in Table 2.

Table 1: Parameter sets for YUANYANG.KEM.

Target security level	128 bits	256 bits	512 bits
ring degree d	512	1024	2048
modulus q	2689	4481	7681
k	3	4	4
σ_f	1.69	1.60	1.49
σ_s	$\sqrt{3}$	$\sqrt{2}$	$\sqrt{2}$
σ_e	$\sqrt{2.5}$	$\sqrt{1.6}$	1
decryption failure rate	$2^{-130.5}$	$2^{-270.2}$	$2^{-531.5}$

4.2 Approximate Gaussian sampling

YUANYANG.KEM uses Gaussian-like secret and encryption randomness. We use 7 bits to sample a half approximate Gaussian using an inverse CDT sampler, and one bit for the sign.

Algorithm 1 $\mathbb{Z}$ -Sampler	
Input	A CDT table T
Output	A random value v
<pre> 1. $v \leftarrow 0$ 2. $r \leftarrow \text{Uniform}([0..255])$ 3. $s \leftarrow 1 - 2(r \% 2)$ 4. $r \leftarrow r \gg 1$ 5. for $0 \leq i < T.\text{length}$ do 6. $v \leftarrow v + (T[i] \leq r)$ return vs </pre>	

The output probability of 0 is therefore $T[0]/128$, and for $i > 0$ and $-i$, it is $\frac{T[i]-T[i-1]}{256}$. For $i = T.\text{length}$ this is replaced by $\frac{128-T[i-1]}{256}$. The concrete values are shown in Table 2.

Table 2: CDT tables used in sampling.

standard deviation	128 bits	256 bits	512 bits
σ_f	[30, 81, 111, 123, 127]	[32, 85, 114, 125]	[34, 89, 117, 126]
σ_s	[30, 79, 109, 123, 127]	[36, 92, 119, 127]	[36, 92, 119, 127]
σ_e	[32, 85, 114, 125]	[40, 99, 122, 127]	[51, 113, 127]

4.3 Underlying Public-Key Encryption

Key generation.

Algorithm 2 samples a short pair $(f, g) \in \mathcal{R} \times \mathcal{R}$ from a discrete Gaussian of standard deviation σ_f and outputs $h = g/f \bmod q$ as the public key. The value of σ_f matches the one used by YUANYANG.DSA, which is what enables the two schemes to share keys. Two invertibility checks ensure that decryption is well defined: f must be invertible both modulo q (used in public key generation) and modulo 2 (used to recover the message bits during decryption). The auxiliary inverse f_2^{-1} is precomputed and stored in the secret key.

Algorithm 2 PKE.KeyGen	
Input	The ring $\mathcal{R} = \mathbb{Z}[X]/(X^d + 1)$, an integer $q \in \mathbb{N}$, and $\sigma_f > 0$
Output	A public key $h \in \mathcal{R}_q$ and a secret key $(f, g, f_2^{-1}) \in \mathcal{R} \times \mathcal{R} \times \mathcal{R}$ such that $h = g/f \bmod q$ and $f \cdot f_2^{-1} = 1 \bmod 2$
1. $f, g \leftarrow D_{\mathcal{R}, \sigma_f}$ 2. if f is not invertible in $\mathcal{R}_2$ then restart 3. if f is not invertible in $\mathcal{R}_q$ then restart 4. Compute $h \leftarrow f^{-1}g \bmod q$ 5. Compute f_2^{-1} such that $f \cdot f_2^{-1} = 1 \bmod 2$ 6. return (pk = h, sk = (f, g, f_2^{-1}))	

Encryption.

The plaintext space is $\{0, 1\}^\lambda$, with $\lambda = d/4$ matching the claimed classical security level. **Encrypt** (Algorithm 3) embeds a fresh random string in the low coefficients of $\mathbb{Z}_2[X]/(X^{d/2} + 1)$, masks it via the small modulus p , adds the fresh randomness term $hs + e$, and then obtains the intermediate ciphertext $\tilde{c}$. The randomness s and e are drawn from the discrete Gaussian distributions $D_{\mathcal{R}, \sigma_s}$ and $D_{\mathcal{R}, \sigma_e}$ seeded by ρ . Finally, YUANYANG.KEM compresses $\tilde{c}$ into a smaller ciphertext c via the coefficient-wise compression function $\text{Compress}: \mathcal{R}_q \rightarrow \mathcal{R}$:

$$\text{Compress}(x) = \left\lfloor \frac{x - 1 + k\%2}{k} \right\rfloor.$$

To make the decryption-failure probability independent of the external message, **Encrypt** also uses a simple ACWC transformation [DHK⁺23]: it encrypts a fresh random string msg' and includes $\text{msg}' \oplus F(\text{msg}')$ as part of the ciphertext, where F is a random oracle. The IND-CPA proof relies on the NTRU assumption for the public key and the Ring-LWE assumption for the mask $hs + e$. The decryption-failure analysis separately accounts for the rounding error introduced by compression.

Algorithm 3 Encrypt	
Input	A public key $\text{pk} = h \in \mathcal{R}_q$, the seed $\rho \in \{0, 1\}^\lambda$, and the message $\text{msg} \in \{0, 1\}^\lambda$
Output	A ciphertext $(c, u) \in \mathcal{R} \times \{0, 1\}^\lambda$
1. $\text{msg}' \leftarrow \\$(\mathcal{U}(\{0, 1\}^\lambda); \rho)$ 2. $m \leftarrow (\text{msg}', 0, \dots, 0) \in \mathbb{Z}_2[X]/(X^{d/2} + 1)$ 3. $s \leftarrow \\$(\mathcal{D}_{\mathcal{R}, \sigma_s}; \rho)$, $e \leftarrow \\$(\mathcal{D}_{\mathcal{R}, \sigma_e}; \rho)$ 4. $\bar{c} \leftarrow (hs + e + mp^{-1}) \bmod q$ $\triangleright p^{-1} = \frac{(q+1)(1-X^{d/2})}{2}$ 5. $c \leftarrow \text{Compress}(\bar{c})$ 6. return $(c, F(\text{msg}') \oplus \text{msg})$	

Decryption.

The decryption **Decrypt** follows the classical NTRU pattern, with one key addition: a lightweight overflow-correction step that recovers from a single overshoot of the small-coefficient assumption. This correction is what makes the decryption failure rate negligible at all parameter sets. For a polynomial $v = \sum_{i=0}^{d/2-1} v_i X^i$ in $\mathbb{Z}[X]/(X^{d/2} + 1)$, we let $\text{Low}(v) = \sum_{i=0}^{d/4-1} v_i X^i$ and $\text{High}(v) = \sum_{i=d/4}^{d/2-1} v_i X^i$.

The procedure is summarized in Algorithm 4. Lines 1–2 mirror the standard NTRU unmasking. Because the encrypted message coordinates have mean one half on the low part, line 3 removes this bias before a centered lift; even k also bias the rounding at compression. The overflow term a is then identified by **ErrCorrection** (Algorithm 5). Removing qa leaves a polynomial that decodes uniquely to the message by multiplication with f_2^{-1} modulo p .

Algorithm 4 Decrypt	
Input	A ciphertext $(c, u) \in \mathcal{R} \times \{0, 1\}^\lambda$, the secret key $\text{sk} = (f, g, f_2^{-1})$
Output	The message $\text{msg} \in \{0, 1\}^\lambda$
1. $\bar{c} \leftarrow k \cdot c \bmod q$ $\triangleright \bar{c} = hs + e + e_{\text{cmp}} + p^{-1}m$ 2. $\bar{c}' \leftarrow \bar{c}fp \bmod q$ 3. $\bar{c}' \leftarrow \bar{c}' - q \left\lfloor \frac{\bar{c}' - \frac{1}{2}f((1-k\%2)(\sum_{i=d/2}^{d-1} 2X^i) + \sum_{i=0}^{d/4-1} X^i)}{q} \right\rfloor$ $\triangleright \bar{c}' = p(gs + f(e + e_{\text{cmp}})) + fm + qa$ 4. $a \leftarrow \text{ErrCorrection}(\bar{c}', \text{sk})$ 5. $C \leftarrow \bar{c}' - qa$ 6. $m \leftarrow C \cdot f_2^{-1} \bmod p$ 7. $\text{msg}' \leftarrow \text{Low}(m)$ 8. return $F(\text{msg}') \oplus u$	

The overflow-correction routine **ErrCorrection** exploits the fact that, with overwhelming probability, the overflow takes the form $a \in \{0\} \cup \{\pm X^i : 0 \leq i < d\}$. Modulo $p = 1 + X^{d/2}$, the candidates with exponents i and $i + d/2$ collapse up to sign, so it is enough to enumerate $0 \leq i < d/2$. The routine selects the unique candidate for which the residue modulo p has zero High-part, as expected for a valid plaintext.

Algorithm 5 ErrCorrection	
Input	A polynomial t promised to have the form $p(gs + fe) + fm + qa$, where $a \in \{0\} \cup \{\pm X^i : 0 \leq i < d\}$ and $m = (\text{msg}, 0, \dots, 0) \in \mathbb{Z}_2[X]/(X^{d/2} + 1)$, together with the secret key $\text{sk} = (f, g, f_2^{-1})$
Output	The overflow error a
1. $\bar{t} \leftarrow (t \bmod p)$ $\triangleright \bar{t} = fm + a \bmod p$ 2. $\bar{t}' \leftarrow \bar{t} \cdot f_2^{-1} \bmod p$ $\triangleright \bar{t}' = m + a \cdot f_2^{-1} \bmod p$ 3. for $a' \in \{0\} \cup \{\pm X^i : 0 \leq i < d/2\}$ do 4. $m' \leftarrow (\bar{t}' - a' \cdot f_2^{-1}) \bmod p$ 5. if $\text{High}(m') = 0$ then return a'	

There are faster algorithms for finding a in Algorithm 5; for example, a Knuth–Morris–Pratt string search runs in linear time. Because that search is not naturally constant-time, we use a different method. A folklore observation is that all values $\langle u, vX^i \rangle$ can be computed efficiently: each is the constant coefficient of uv^*X^i , where v^* is the conjugate of v . If u and v are random binary elements, then $\langle u - \text{Low}(u), v \rangle = \langle u - \text{Low}(u), u - \text{Low}(u) \rangle$ occurs with high probability only when $\text{High}(u - v) = 0$. It follows that we can identify any overflow equal to $\pm X^i$ for $0 \leq i < d/4$ by computing $\bar{t} = t \bmod p \in \mathbb{Z}[X]/(X^{d/2} + 1)$, $\bar{f} = f_2^{-1} \bmod p$, and checking the coefficients of

$$(\bar{t} - \text{Low}(\bar{t}))^* \bar{f} \in \mathbb{Z}[X]/(X^{d/2} + 1).$$

Larger values of i lead to negative coefficients in $\text{High}(\bar{f}X^i)$, so they are checked by computing $\bar{f}' = \bar{f}X^{d/4} \bmod 2$ and then using

$$(\bar{t} - \text{Low}(\bar{t}))^* \bar{f}' \in \mathbb{Z}[X]/(X^{d/2} + 1).$$

Equivalently, one can use a single multiplication modulo $X^{d/2} - 1$; Algorithm 6 implements this idea in constant time.

Algorithm 6 ErrCorrection	
Input	A polynomial t promised to have the form $p(gs + fe) + fm + qa$, where $a \in \{0\} \cup \{\pm X^i : 0 \leq i < d\}$ and $m = (\text{msg}, 0, \dots, 0) \in \mathbb{Z}_2[X]/(X^{d/2} + 1)$, together with the secret key $\text{sk} = (f, g, f_2^{-1})$
Output	The overflow error a
1. $\bar{t} \leftarrow t \bmod (p, 2)$ $\triangleright \bar{t} = fm + a \bmod (p, 2)$ 2. $\bar{t}' \leftarrow \bar{t} \cdot f_2^{-1} \bmod (p, 2)$ $\triangleright \bar{t}' = m + a \cdot f_2^{-1} \bmod (p, 2)$ 3. $u \leftarrow \bar{t}' - \text{Low}(\bar{t}')$ 4. $v \leftarrow u \cdot (f_2^{-1} \bmod (p, 2))$ 5. $v \leftarrow v \bmod X^{d/2} - 1$ $\triangleright \sum_{i=0}^{d/2-1} \langle u, (f_2^{-1} \bmod (p, 2))X^i \bmod X^{d/2} - 1 \rangle X^i$ 6. $a' \leftarrow 0$ 7. for $0 \leq i < d/2$ do 8. $a' \leftarrow a' + X^i \cdot (v[i] == \text{Hamming}(u))$ $\triangleright \text{Hamming}(u) = \ u\ ^2$	
return $a' \cdot (\text{Hamming}(u) > 0)$	

4.4 KEM Construction

The KEM is obtained from the underlying PKE by a Fujisaki–Okamoto-style transform. Key generation (Algorithm 7) wraps PKE.KeyGen with a fallback key K' used during implicit rejection and caches the public-key hash $\mathbf{H}_{\text{pk}}$ for later reuse. Encapsulation (Algorithm 8) commits to a uniform message, derives the encryption seed and the pre-key from a hash of the message and the public key, and re-encrypts deterministically. Decapsulation (Algorithm 9) recomputes the ciphertext from the decoded message and accepts only if the re-encryption matches; otherwise it returns an unforgeable but unrelated key derived from K' .

Algorithm 7 KeyGen
Output The public key pk and the private key sk
1. $(\text{pk}, \text{sk}') \leftarrow \text{PKE.KeyGen}$ 2. $K' \leftarrow \mathcal{U}(\{0, 1\}^\lambda)$ 3. $\mathbf{H}_{\text{pk}} \leftarrow \text{Hash}_{\text{pk}}(\text{pk})$ 4. $\text{sk} \leftarrow (K', \text{sk}', \text{pk}, \mathbf{H}_{\text{pk}})$ 5. return (pk, sk)

Algorithm 8 Encapsulate
Input The public key pk Output The ciphertext c and the shared key K
1. $\text{msg} \leftarrow \mathcal{U}(\{0, 1\}^\lambda)$ 2. $(\bar{K}, \rho) \leftarrow \text{Hash}_{\text{msg}}(\text{msg}, \text{Hash}_{\text{pk}}(\text{pk}))$ 3. $c \leftarrow \text{Encrypt}(\text{pk}, \rho, \text{msg})$ 4. $K \leftarrow \text{Hash}_K(\bar{K}, c)$ 5. return (c, K)

Algorithm 9 Decapsulate
Input The ciphertext c and the private key sk Output The shared key K
1. Parse sk as $(K', \text{sk}', \text{pk}, \mathbf{H}_{\text{pk}})$ 2. $\text{msg} \leftarrow \text{Decrypt}(c, \text{sk}')$ 3. $(\bar{K}, \rho) \leftarrow \text{Hash}_{\text{msg}}(\text{msg}, \mathbf{H}_{\text{pk}})$ 4. if $c = \text{Encrypt}(\text{pk}, \rho, \text{msg})$ then 5. $K \leftarrow \text{Hash}_K(\bar{K}, c)$ 6. else 7. $K \leftarrow \text{Hash}_K(K', c)$ 8. return K

Remark 4.1. There are several random oracles used in YUANYANG.KEM, including F in **Encrypt**, Hash_{pk} in **KeyGen**, Hash_{msg} and Hash_K in **Encapsulate**. We use different prefixes to separate RO domains.

5 Security Statements and Analyses

This section states the hardness assumptions used by YUANYANG.KEM, proves the IND-CPA security of the underlying PKE, recalls the Fujisaki–Okamoto (FO) transformation, and provides the concrete security and decryption-failure estimates for the proposed parameter sets.

5.1 Hardness Assumptions

All assumptions are stated over the ring $\mathcal{R} = \mathbb{Z}[X]/(X^d + 1)$ and its quotient $\mathcal{R}_q = \mathcal{R}/q\mathcal{R}$ used in Section 4. The key-generation distribution samples $f, g \leftarrow D_{\mathcal{R}, \sigma_f}$ and conditions on the invertibility tests required by Algorithm 2; in particular, f is invertible in $\mathcal{R}_q$ and in $\mathcal{R}_2$. The encryption distributions sample $s \leftarrow D_{\mathcal{R}, \sigma_s}$ and $e \leftarrow D_{\mathcal{R}, \sigma_e}$ as in Algorithm 3.

Definition 5.1 (Decisional NTRU assumption). Let $f, g \leftarrow D_{\mathcal{R}, \sigma_f}$ be sampled as in Algorithm 2, and let

$$h = f^{-1}g \bmod q \in \mathcal{R}_q.$$

Let $u \leftarrow \mathcal{U}(\mathcal{R}_q)$ be uniform and independent. The decisional NTRU assumption states that the NTRU public-key distribution $h = f^{-1}g \bmod q$ is computationally indistinguishable from the uniform distribution over $\mathcal{R}_q$. For an adversary $\mathcal{B}$, its distinguishing advantage is

$$\text{Adv}_{\mathcal{R}, q, \sigma_f}^{\text{NTRU}}(\mathcal{B}) = \left| \Pr[\mathcal{B}(f^{-1}g \bmod q) = 1] - \Pr[\mathcal{B}(u) = 1] \right|.$$

Definition 5.2 (Decisional Ring-LWE assumption). Let $a \leftarrow \mathcal{U}(\mathcal{R}_q)$, $s \leftarrow D_{\mathcal{R}, \sigma_s}$, $e \leftarrow D_{\mathcal{R}, \sigma_e}$, and $u \leftarrow \mathcal{U}(\mathcal{R}_q)$ be independent. The decisional Ring-LWE assumption states that the two distributions

$$(a, as + e) \quad \text{and} \quad (a, u)$$

are computationally indistinguishable. For an adversary $\mathcal{C}$, its distinguishing advantage is

$$\text{Adv}_{\mathcal{R}, q, \sigma_s, \sigma_e}^{\text{RLWE}}(\mathcal{C}) = \left| \Pr[\mathcal{C}(a, as + e) = 1] - \Pr[\mathcal{C}(a, u) = 1] \right|.$$

5.2 PKE Security

Let $\Pi_{\text{PKE}} = (\text{PKE}, \text{KeyGen}, \text{Encrypt}, \text{Decrypt})$ denote the public-key encryption scheme specified in Algorithms 2, 3, and 4. The function $F : \{0, 1\}^\lambda \rightarrow \{0, 1\}^\lambda$ used in Algorithm 3 is modeled as a random oracle. We first spell out the form of a challenge encryption. On input a public key $\text{pk} = h$ and a message $\text{msg} \in \{0, 1\}^\lambda$, Algorithm 3 samples

$$\text{msg}' \leftarrow \mathcal{U}(\{0, 1\}^\lambda), \quad m = (\text{msg}', 0, \dots, 0) \in \mathbb{Z}_2[X]/(X^{d/2} + 1),$$

samples $s \leftarrow D_{\mathcal{R}, \sigma_s}$ and $e \leftarrow D_{\mathcal{R}, \sigma_e}$, computes

$$\tilde{c} = (hs + e + mp^{-1}) \bmod q, \quad c = \text{Compress}(\tilde{c}),$$

and outputs

$$(c, F(\text{msg}') \oplus \text{msg}).$$

Thus the NTRU component encrypts the fresh random string msg' , while the external message is hidden by the random-oracle value $F(\text{msg}')$.

Theorem 5.3 (IND-CPA security of YUANYANG.PKE in the ROM). For every IND-CPA adversary $\mathcal{A}$ against Π_{PKE} that makes at most q_F classical queries to F , there exist adversaries $\mathcal{B}$ and $\mathcal{C}$, running in essentially the same time as $\mathcal{A}$, such that

$$\text{Adv}_{\Pi_{\text{PKE}}}^{\text{IND-CPA}}(\mathcal{A}) \leq \text{Adv}_{\mathcal{R}, q, \sigma_f}^{\text{NTRU}}(\mathcal{B}) + \text{Adv}_{\mathcal{R}, q, \sigma_s, \sigma_e}^{\text{RLWE}}(\mathcal{C}) + q_F 2^{-\lambda}.$$

Consequently, if the NTRU and Ring-LWE assumptions in Section 5.1 hold and λ is the message length used in Algorithm 3, then YUANYANG.PKE is IND-CPA secure in the random-oracle model.

Proof. Let W_i be the event that $\mathcal{A}$ wins in Game i .

Game 0. Game 0 is the standard IND-CPA experiment for Π_{PKE} . The challenger generates (pk, sk) by Algorithm 2, gives $\text{pk} = h$ to $\mathcal{A}$, receives two messages $\text{msg}_0, \text{msg}_1 \in \{0, 1\}^\lambda$, samples $b \leftarrow \{0, 1\}$, and computes the challenge ciphertext by Algorithm 3. Equivalently, the challenge ciphertext is

$$(c^*, u^*) = (\text{Compress}((hs + e + mp^{-1}) \bmod q), F(\text{msg}'_b) \oplus \text{msg}_b),$$

where

$$\text{msg}' \leftarrow \mathcal{U}(\{0, 1\}^\lambda), \quad m = (\text{msg}', 0, \dots, 0), \quad s \leftarrow D_{\mathcal{R}, \sigma_s}, \quad e \leftarrow D_{\mathcal{R}, \sigma_e}.$$

By definition,

$$\text{Adv}_{\Pi_{\text{PKE}}}^{\text{IND-CPA}}(\mathcal{A}) = \left| \Pr[W_0] - \frac{1}{2} \right|.$$

Game 1. Game 1 is identical to Game 0 except that the public key is sampled as

$$h \leftarrow \mathcal{U}(\mathcal{R}_q)$$

instead of $h = f^{-1}g \bmod q$. This is exactly the decisional NTRU replacement from the assumption above. Hence there exists an adversary $\mathcal{B}$ such that

$$|\Pr[W_0] - \Pr[W_1]| \leq \text{Adv}_{\mathcal{R}, q, \sigma_f}^{\text{NTRU}}(\mathcal{B}).$$

Game 2. Game 2 is identical to Game 1 except that the Ring-LWE mask $hs + e$ in the challenge ciphertext is replaced by an independently uniform element of $\mathcal{R}_q$. Thus the challenge ciphertext is now computed as

$$(c^*, u^*) = (\text{Compress}((z + mp^{-1}) \bmod q), F(\text{msg}') \oplus \text{msg}_b), \quad z \leftarrow \mathcal{U}(\mathcal{R}_q).$$

This is the decisional Ring-LWE replacement from the assumption above: in Game 1 the pair $(h, hs + e)$ is distributed as $(a, as + e)$, while in Game 2 it is distributed as (a, z) with $a, z \leftarrow \mathcal{U}(\mathcal{R}_q)$. Therefore there exists an adversary $\mathcal{C}$ such that

$$|\Pr[W_1] - \Pr[W_2]| \leq \text{Adv}_{\mathcal{R}, q, \sigma_s, \sigma_e}^{\text{RLWE}}(\mathcal{C}).$$

Game 3. Game 3 is identical to Game 2 except that the second component of the challenge ciphertext is sampled uniformly and independently:

$$(c^*, u^*) = (\text{Compress}((z + mp^{-1}) \bmod q), u^*), \quad z \leftarrow \mathcal{U}(\mathcal{R}_q), \quad u^* \leftarrow \mathcal{U}(\{0, 1\}^\lambda).$$

In Game 3, the value $z + mp^{-1}$ is uniform over $\mathcal{R}_q$ for every fixed m , because z is uniform over $\mathcal{R}_q$. Hence $c^* = \text{Compress}((z + mp^{-1}) \bmod q)$ is independent of msg' and of the challenge bit b . The second component u^* is also uniform and independent of b . Therefore the whole challenge ciphertext is independent of b , and

$$\Pr[W_3] = \frac{1}{2}.$$

It remains to bound the transition from Game 2 to Game 3. Let *Query* be the event that $\mathcal{A}$ queries the random oracle F at the point msg' used in the challenge ciphertext. The two games are identical until *Query* occurs: if $\mathcal{A}$ never queries F at msg' , then the value $F(\text{msg}')$ used in $F(\text{msg}') \oplus \text{msg}_b$ is a fresh uniform λ -bit string hidden from $\mathcal{A}$, and therefore the second component of the challenge ciphertext is distributed exactly as a uniform element of $\{0, 1\}^\lambda$. Thus

$$|\Pr[W_2] - \Pr[W_3]| \leq \Pr[\text{Query}].$$

Moreover, in Game 2 the first component c^* is independent of msg' , as argued above. Therefore each classical query to F hits the hidden value $\text{msg}' \leftarrow \mathcal{U}(\{0, 1\}^\lambda)$ with probability at most $2^{-\lambda}$, even conditioned on the adversary's current view. Since $\mathcal{A}$ makes at most q_F queries to F , the union bound gives

$$\Pr[\text{Query}] \leq q_F 2^{-\lambda}.$$

Combining the game hops, we obtain

$$\begin{aligned} \text{Adv}_{\Pi_{\text{PKE}}}^{\text{IND-CPA}}(\mathcal{A}) &= \left| \Pr[W_0] - \frac{1}{2} \right| \\ &\leq |\Pr[W_0] - \Pr[W_1]| + |\Pr[W_1] - \Pr[W_2]| + |\Pr[W_2] - \Pr[W_3]| \\ &\leq \text{Adv}_{\mathcal{R}, q, \sigma_f}^{\text{NTRU}}(\mathcal{B}) + \text{Adv}_{\mathcal{R}, q, \sigma_s, \sigma_e}^{\text{RLWE}}(\mathcal{C}) + q_F 2^{-\lambda}. \end{aligned}$$

This proves the theorem.

5.3 KEM Security

The transformation from YuanYang.PKE to YuanYang.KEM follows the FO transformation with implicit rejection and public-key prefix hashing [HHK17, JZC⁺18, DHK⁺21].

Theorem 5.4 (FO transform with implicit rejection, ROM form). *Let Π_{PKE} be an IND-CPA-secure public-key encryption scheme with message space $\mathcal{M}$ and decryption error at most δ . Let Π_{KEM} be the KEM obtained from Π_{PKE} by the FO transform with implicit rejection and public-key prefix hashing, as in Algorithms 7, 8, and 9. Let λ be the bit-length of K' sampled in Algorithm 7. Then, for every classical IND-CCA adversary $\mathcal{A}$ making at most q_F random-oracle queries to the FO random oracles and at most q_D decapsulation queries, there exists an IND-CPA adversary $\mathcal{B}$ against Π_{PKE} , with essentially the same running time, such that*

$$\text{Adv}_{\Pi_{\text{KEM}}}^{\text{IND-CCA}}(\mathcal{A}) \leq 2\text{Adv}_{\Pi_{\text{PKE}}}^{\text{IND-CPA}}(\mathcal{B}) + \frac{2(q_F + 1)}{|\mathcal{M}|} + \frac{q_F}{2^\lambda} + (q_F + q_D)\delta.$$

In particular, if Π_{PKE} is IND-CPA secure, δ is negligible, $|\mathcal{M}|$ is superpolynomial in the security parameter, and $2^{-\lambda}$ is negligible, then Π_{KEM} is IND-CCA secure in the classical random-oracle model.

Theorem 5.5 (FO transform with implicit rejection, QROM form). *Let Π_{PKE} be an IND-CPA-secure public-key encryption scheme with message space $\mathcal{M}$ and decryption error at most δ . Let Π_{KEM} be the KEM obtained from Π_{PKE} by the FO transform with implicit rejection and public-key prefix hashing, as in Algorithms 7, 8, and 9. Let λ be the bit-length of K' sampled in Algorithm 7. Then, for every quantum IND-CCA adversary $\mathcal{A}$ making at most q_F quantum random-oracle queries to the FO random oracles and at most q_D decapsulation queries, there exists a quantum IND-CPA adversary $\mathcal{B}$ against Π_{PKE} , with essentially the same running time, such that, for $q := q_F + q_D + 1$,*

$$\text{Adv}_{\Pi_{\text{KEM}}}^{\text{IND-CCA}}(\mathcal{A}) \leq 2\sqrt{q\text{Adv}_{\Pi_{\text{PKE}}}^{\text{IND-CPA}}(\mathcal{B})} + \frac{4q}{\sqrt{|\mathcal{M}|}} + \frac{4(q_F + 1)}{2^{\lambda/2}} + 16q^2\delta + \frac{1}{|\mathcal{M}|}.$$

In particular, if Π_{PKE} is IND-CPA secure, δ is negligible, $|\mathcal{M}|$ is superpolynomial in the security parameter, and $2^{-\lambda}$ is negligible, then Π_{KEM} is IND-CCA secure in the quantum random-oracle model.

Together with Theorem 5.3, these bounds give the asymptotic IND-CCA security of YuanYang.KEM from the NTRU and Ring-LWE assumptions, the FO transform with implicit rejection, and the decryption-error bound used in the FO theorem. The concrete parameter choices below are therefore driven by direct cryptanalytic estimates and decryption-failure estimates; the FO reduction is used as a qualitative security justification rather than as a tight concrete-security estimator.

5.4 Concrete Security

The reduction above provides qualitative assurance that breaking YuanYang.KEM is at least as hard as solving NTRU and RLWE, and hence rests on well-studied lattice assumptions. As is now standard practice for lattice-based signatures and KEMs – including the NIST standards ML-DSA, ML-KEM, and FN-DSA – we do *not* use the reduction itself to drive the choice of parameters. Provable bounds in this regime tend to be loose, and tracking them strictly would force unnecessarily large parameters. Instead, we follow the now-standard methodology of selecting parameters so that the cost of the best known cryptanalytic attacks comfortably exceeds the targeted security level. The remainder of this section is therefore devoted to estimating that cost: we describe the model used for lattice reduction, then analyze the two attack avenues – NTRU attack and RLWE attack – that determine our parameter choices.

5.4.1 Model for lattice reduction.

Throughout, we adopt the *Geometric Series Assumption* (GSA), which states that the Gram–Schmidt norms of a reduced basis decay geometrically. Concretely, for the self-dual BKZ (DBKZ) reduction algorithm of Micciancio and Walter [MW17], the output basis $(\mathbf{b}_1, \dots, \mathbf{b}_N)$ produced with block size β on a lattice $\mathcal{L}$ of rank N satisfies

$$\|\mathbf{b}_i^*\| = \delta_\beta^{N+1-2i} \det(\mathcal{L})^{1/N}, \quad \text{where } \delta_\beta = \left(\frac{(\pi\beta)^{1/\beta} \cdot \beta}{2\pi e} \right)^{1/(2(\beta-1))},$$

and $\mathbf{b}_i^*$ denotes the i -th Gram–Schmidt vector of the basis.

5.4.2 NTRU attack.

The NTRU attack consists in retrieving the secret key $(f, g) \in \mathcal{R}^2$ from the public data q and h alone. To the best of our knowledge, the strongest attacks proceed via lattice reduction. They build the algebraic lattice over $\mathcal{R}$ spanned by $(q, 0)$ and $(h, 1)$ – the public NTRU basis – and search for the lattice vector $\mathbf{s} = (\mathbf{g}, \mathbf{f})$ (or any functionally equivalent vector such as $(\mu g, \mu f)$ for a unit μ of the number field) of norm $\|\mathbf{s}\| = \sqrt{2d}\sigma$.

To avoid enumerating the full sphere of radius $\sqrt{2d}\sigma$ – which, by the Gaussian heuristic, contains roughly $(2d\sigma^2/q)^d$ vectors – we rely on the so-called *projection trick*. Concretely, fix a block size β and run DBKZ on the public basis, yielding $[\mathbf{b}_1, \dots, \mathbf{b}_{2d}]$. If the *projection* of the secret key onto $\mathcal{P}$, the orthogonal complement of $\text{Span}(\mathbf{b}_1, \dots, \mathbf{b}_{2d-\beta})$, can be recovered, then the full secret key is obtained in polynomial time via Babai’s nearest-plane algorithm. It therefore suffices to find the projected secret key among the shortest vectors of the lattice generated by the last β basis vectors projected onto $\mathcal{P}$. Sieving on this projected lattice classically recovers all vectors of norm at most $\sqrt{4/3}\ell$, where ℓ is the norm of the first Gram–Schmidt vector in the projected block, namely $\mathbf{b}_{2d-\beta+1}^*$. Under the GSA,

$$\ell = \sqrt{q} \delta_\beta^{-2d+2\beta-1}.$$

Modeling $\mathbf{s}$ as a random vector of norm $\sqrt{2d}\sigma$ and using the GSA to bound the Gram–Schmidt norms of $\mathbf{b}_1^*, \dots, \mathbf{b}_{2d-\beta}^*$, the projection of $\mathbf{s}$ onto $\mathcal{P}$ has norm roughly

$$\sqrt{\frac{\beta}{2d}} \|\mathbf{s}\| = \beta^{1/2} \sigma.$$

The projection is therefore recovered by sieving as soon as $\beta^{1/2} \sigma \leq \sqrt{4/3}\ell$, i.e. when

$$\sigma^2 \leq \frac{4q}{3\beta} \delta_\beta^{-4d+4\beta-2}. \quad (1)$$

5.4.3 RLWE attack.

We evaluate the Ring-LWE after coefficient embedding as an asymmetric LWE instance

$$(\mathbf{A}, \mathbf{b} = \mathbf{A}\mathbf{s} + \mathbf{e}) \in \mathbb{Z}_q^{d \times d} \times \mathbb{Z}_q^d, \quad \mathbf{s} \leftarrow D_{\mathbb{Z}^d, \sigma_s}, \quad \mathbf{e} \leftarrow D_{\mathbb{Z}^d, \sigma_e},$$

where the secret and error widths σ_s and σ_e are not necessarily equal. The main known lattice attacks against this instance are primal and dual attacks.

Primal attack. The primal attack embeds the RLWE relation into a unique-SVP instance. Equivalently, the attacker looks for the short vector determined by $(\mathbf{s}, \mathbf{e})$. For this asymmetric setting, the useful refinement is to rescale the secret coordinates by $c = \frac{\sigma_e}{\sigma_s}$, so that the embedded target is balanced as $(c\mathbf{s}, \mathbf{e}, \sigma_e)$ and has expected norm about $\sigma_e \sqrt{2d+1}$. The primal estimate is obtained by optimizing the BKZ block size for this rescaled embedded lattice and applying the same Core-SVP cost model as for the NTRU attack.

Dual attack. The dual attack searches for short relations that turn the public RLWE sample into a distinguishing problem. One splits the coefficient vector of the secret as $\mathbf{s} = (\mathbf{s}_0, \mathbf{s}_1)$ and searches for several short relations $\mathbf{v}_i$ whose action on the $\mathbf{s}_0$ -part is a short vector $\mathbf{w}_i$ modulo q . For each such relation, the sample satisfies a congruence of the form

$$\langle \mathbf{v}_i, \mathbf{b} \rangle \equiv \langle \mathbf{u}_i, \mathbf{s}_1 \rangle + \langle \mathbf{w}_i, \mathbf{s}_0 \rangle + \langle \mathbf{v}_i, \mathbf{e} \rangle \pmod{q},$$

where $\mathbf{u}_i$ is public and determined by the same relation. The attacker enumerates the smaller part $\mathbf{s}_1$ and subtracts the corresponding public term. For the correct guess, the residual is no longer uniform modulo q but is concentrated with width roughly $\sqrt{\sigma_s^2 \|\mathbf{w}_i\|^2 + \sigma_e^2 \|\mathbf{v}_i\|^2}$. The estimated cost is the minimum, over the split size and BKZ block size, of the lattice-reduction cost for producing enough such relations plus the enumeration-and-testing cost for $\mathbf{s}_1$. In the parameter selection, the reported RLWE security level is the minimum of the best primal and dual estimates under this asymmetric model.

5.4.4 Concrete security estimates.

We assess the concrete security of YUANYANG.KEM by following the standard cryptanalytic methodology: estimating the costs of the best known NTRU and RLWE attacks.

These cost estimates are converted into concrete bit-security values via the methodology of NEWHOPE [ADPS16], often called the “core-SVP methodology”. In this model [BDGL16, Laa16], the bit complexity of lattice sieving – asymptotically the best SVP oracle – in dimension β is set to $\lfloor 0.292\beta \rfloor$ in the classical setting and $\lfloor 0.265\beta \rfloor$ in the quantum setting.

Table 3: Concrete security estimates for YUANYANG.KEM (C is classical security, Q is quantum security.)

	YUANYANG.KEM-512	YUANYANG.KEM-1024	YUANYANG.KEM-2048
Key recovery (C/Q)	129/117	267/243	546/496
Message recovery (C/Q)	128/116	256/232	520/472

5.4.5 Other attacks.

For completeness, we review other attacks that have been considered against NTRU-style or LWE-style schemes. None of them improves on the estimates reported in Table 3 for the proposed parameter sets.

Hybrid attack. The hybrid attack combines lattice reduction with a meet-in-the-middle search and was introduced as an improved attack against early NTRU [HG07], with later refinements in [BGPW16, Ngu21, SC19]. It is effective only against NTRU or LWE instances with very sparse secret or error vectors. The secrets and errors in YUANYANG.KEM carry enough entropy, with coefficient standard deviations at least 1, to neutralize this approach.

Dual attack. For NTRU key recovery, the classical dual attack is not the leading attack and does not improve on the projection-based primal analysis above. For the Ring-LWE message-recovery instance, dual attacks are already included in the estimator comparison leading to Table 3, and they do not lower the reported concrete security levels.

Overstretched NTRU attacks. The subfield and overstretched-modulus attacks of [ABD16, KF17] apply only when q is far larger than d . The moduli used in YUANYANG.KEM are well below this regime, so these attacks do not apply to the proposed parameters.

Algebraic attacks. YUANYANG.KEM uses a power-of-two cyclotomic ring, which gives the system a rich algebraic structure. Known algebraic-lattice speed-ups, such as those in [KEF20], improve lattice reduction by at most polynomial factors and do not affect the Core-SVP security levels reported above.

5.5 Decryption Failure Analysis

The ACWC transformation [DHK⁺23] makes the decryption-failure probability independent of the external message encrypted by the KEM. For an honest ciphertext, after decompression we write

$$\text{Decompress}(c) = hs + e + e_{\text{cmp}} + p^{-1}m,$$

where e_{cmp} denotes the rounded error. The intermediate polynomial $\tilde{c}'$ in decryption has the form

$$\tilde{c}' = p(gs + f(e + e_{\text{cmp}})) + fm + qa$$

for an overflow polynomial a . The overflow-correction step is designed to handle the case where $a \in \{0\} \cup \{\pm X^i : 0 \leq i < d\}$. We model the coefficients of

$$p(gs + f(e + e_{\text{cmp}})) + fm$$

as independent centered Gaussian variables of standard deviation

$$\sigma_{\text{dec}} = \sigma_f \sqrt{d} \cdot \sqrt{2\sigma_s^2 + 2\sigma_e^2 + 2\sigma_{\text{cmp}}^2 + \left(\frac{1}{4}\right)^2}$$

where $\sigma_{\text{cmp}} = \sqrt{(k^2 - 1)/12}$ denotes the standard deviation of e_{cmp} and $1/4$ is the standard-deviation contribution of the centered binary message component. Let

$$\tau = \frac{q - 1}{2\sigma_{\text{dec}}}$$

and let P_{est} denote the probability that at least two coefficients exceed the centered modulus boundary. We use the conservative estimate

$$\log_2 P_{\text{est}} = 2 \log_2 \left(2 \operatorname{erfc} \left(\frac{\tau}{\sqrt{2}} \right) \right) + 2 \log_2 d.$$

The extra factor 2 inside the logarithm is a safety margin on the one-coordinate Gaussian tail. This estimates the dominant event that the overflow lies outside the one-error-correctable set.

On the independence hypothesis. Recent reports have suggested strong deviations from the model of independent identically distributed coordinates, with significant amplification of outlier probabilities and therefore larger decryption-failure rates. We sampled 2^{38} ternary ring elements in $\mathbb{Z}[X]/(X^{512} + 1)$ and computed the statistics of the second largest coordinate of ab by absolute value. The element a is fixed, with 363 non-zero values. We found 6 cases beyond 90, 211 samples beyond 85, and 6832 samples beyond 80; under the independence hypothesis, the predicted averages are respectively 4, 173, and 5888 samples. We observe outliers do not deviate much from the independence hypothesis. Furthermore, decryption uses the sum of two independent products of similar size, so we expect even less correlation.

5.6 Joint Security

The defining feature of YuanYang is that the KEM and the signature scheme are deployed with one NTRU public key and one shared NTRU secret core. This section records the resulting joint-security guarantee. The goal is not to repeat the standalone proofs: the IND-CCA proof of YUANYANG.KEM is given above, and the UF-CMA proof of YUANYANG.DSA is given in the companion signature specification. Instead, we make explicit why opening both interfaces under the same key does not create an additional attack surface beyond the losses already accounted for in the two standalone analyses.

For a concrete YuanYang parameter set, a joint key pair is generated once. The public key is the NTRU element

$$\text{pk} = h = g/f \bmod q \in \mathcal{R}_q.$$

The shared secret core contains the short NTRU material used by both algorithms. On the signature side this material is completed into the NTRU trapdoor basis and accompanied by the sampler precomputation needed by YUANYANG.DSA; on the KEM side it is accompanied by the inverse modulo the masking ideal and the FO implicit-rejection state needed by YUANYANG.KEM. These auxiliary values are component-specific caches or rejection states. They do not constitute a second independently generated NTRU key.

Joint KEM security. The joint IND-CCA experiment is the usual KEM IND-CCA experiment, with an additional signing oracle for YUANYANG.DSA under the same public key h . More precisely, after joint key generation, the challenger gives h to the adversary, answers signing queries $M \mapsto \text{Sign}(\text{sk}_{\text{DSA}}, M)$, and answers decapsulation queries $C \mapsto \text{Decaps}(\text{sk}_{\text{KEM}}, C)$. The challenge is the usual KEM challenge: the challenger computes an honest encapsulation (C^*, K_0) , samples an independent key K_1 from the key space, gives (C^*, K_b) for a hidden bit b , and refuses the decapsulation query C^* . The adversary wins if it outputs $b' = b$. We write

$$\text{Adv}_{\text{YY.KEM}}^{\text{J-IND-CCA}}(\mathcal{A}) = \left| \Pr[b' = b] - \frac{1}{2} \right|.$$

This is the security notion relevant when an attacker can see signatures and mount chosen-ciphertext attacks at the same time.

Joint signature security. The joint UF-CMA experiment is the ordinary UF-CMA experiment for YUANYANG.DSA, except that the adversary is additionally given access to the KEM decapsulation oracle under the same public key h . The challenger answers signing queries and decapsulation queries, and the adversary wins if it outputs a pair (M^*, σ^*) such that M^* was not previously submitted to the signing oracle and

$$\text{Verif}(h, M^*, \sigma^*) = 1.$$

We denote this winning probability by

$$\text{Adv}_{\text{YY.DSA}}^{\text{J-UF-CMA}}(\mathcal{A}) = \Pr[\mathcal{A} \text{ wins the joint UF-CMA experiment}].$$

YuanYang is jointly secure if both advantages are negligible for every efficient adversary.

Theorem 5.6 (Joint security of YuanYang). *Assume that all hash and XOF calls are domain-separated as specified in Section 4, and that the decapsulation implementation reveals only the final shared key and no intermediate decoding, correction, or failure information. Then, for every efficient joint adversary $\mathcal{A}$, there exist adversaries $\mathcal{B}_{\text{KEM}}$ against the standalone IND-CCA security of YUANYANG.KEM and $\mathcal{B}_{\text{DSA}}$ against the standalone UF-CMA security of YUANYANG.DSA, with comparable running time and query complexity, such that*

$$\text{Adv}_{\text{YY.KEM}}^{\text{J-IND-CCA}}(\mathcal{A}) \leq \text{Adv}_{\text{YY.KEM}}^{\text{IND-CCA}}(\mathcal{B}_{\text{KEM}}) + \epsilon_{\text{sig-sim}},$$

and

$$\text{Adv}_{\text{YY.DSA}}^{\text{J-UF-CMA}}(\mathcal{A}) \leq \text{Adv}_{\text{YY.DSA}}^{\text{UF-CMA}}(\mathcal{B}_{\text{DSA}}) + \epsilon_{\text{fo-sim}}.$$

Here $\epsilon_{\text{sig-sim}}$ is the signing-simulation loss bounded in the proof of YUANYANG.DSA: the rejection-budget term, the salt-collision term, and the Rényi losses for replacing real trapdoor sampling by the simulated hash-and-sign distribution. Likewise, $\epsilon_{\text{fo-sim}}$ is the decapsulation-simulation loss bounded in the FO analysis of YUANYANG.KEM: the random-oracle terms and the contribution of the decryption-error probability. Consequently, combining the standalone bounds gives joint security under the same NTRU, Ring-LWE, and Ring-ISIS assumptions used by the two standalone schemes.

Proof (Proof overview). We explain the two reductions at the level needed for a specification document. The full game transitions are the same as in the standalone proofs; the only additional point is that the two oracle interfaces are now available under the same h .

First consider the joint IND-CCA experiment for the KEM. The adversary receives a signing oracle in addition to the usual KEM challenge and decapsulation oracle. By the hash-and-sign simulation used in the proof of YUANYANG.DSA, real signatures can be replaced by simulated signatures whose distribution is indistinguishable from the real one up to the loss $\epsilon_{\text{sig-sim}}$. This replacement is possible without knowing the signing trapdoor, because the random-oracle output in the DSA domain is programmed together with a short preimage distributed as in the signature proof. Domain separation is the reason this programming does not alter the random oracles used by the KEM transform. After this replacement, the signing oracle is independent of the KEM challenge except through the public key h , and the remaining experiment is precisely the standalone IND-CCA experiment for YUANYANG.KEM. Any adversary distinguishing the challenge key can therefore be converted into $\mathcal{B}_{\text{KEM}}$, plus the signing-simulation loss.

Now consider the joint UF-CMA experiment for the signature. The extra oracle is KEM decapsulation. In the FO transform with implicit rejection, invalid ciphertexts are answered by a key derived from the fallback secret and the ciphertext, while valid ciphertexts are tied to the unique message recovered by deterministic re-encryption. The standalone KEM proof already shows how to simulate this interface up to the loss governed by random-oracle queries and the decryption-error probability. Since the KEM hash domains are disjoint from the DSA hash-to-target domain, this simulation does not interfere with the programming steps in the signature proof. After replacing the decapsulation oracle by its FO simulation, the adversary sees exactly the same signing interface as in the standalone UF-CMA experiment. A successful fresh forgery therefore yields the same Ring-ISIS solution as in the proof of YUANYANG.DSA, except for the FO-simulation loss $\epsilon_{\text{fo-sim}}$.

The theorem follows by combining these two hybrid arguments. The shared NTRU public key h is already the public input in both standalone analyses, so sharing it does not by itself add a new secret-dependent transcript. The component-specific auxiliary states are also harmless: the signing sampler cache only affects how signatures are produced, and the KEM fallback state only affects implicit rejection. Both effects are already covered by the corresponding standalone simulations. What must be enforced

at implementation level is that the decapsulation routine returns only the final key and that the logical random-oracle domains remain separated by their fixed public labels.

Remark 5.7. In the algorithms above, we write the hash and XOF calls as Hash_{msg} , Hash_K , and Hash_{pk} to keep the description readable. In the standalone security analysis, these calls are modeled as random oracles. For the joint-security statement in this section, we additionally require the random-oracle domains used by YUANYANG.KEM to be separated from the hash domain used by the companion YUANYANG.DSA. Thus, when YuanYang is deployed in the key-reuse setting, the implementation should realize the KEM hash/XOF calls and the signing hash call by using the same underlying XOF only after prepending fixed public labels identifying their respective roles.

6 Performance Evaluation

Integers are encoded with a batch size B of 4 in the public key and the ciphertext of the 256 bit version. Public-key and ciphertext sizes are computed analytically as $\lceil B \log_2 q \rceil \cdot d/8/B$ and $\lceil B \log_2(q/k+2) \rceil \cdot d/(8k)/B + \lambda/8$ bytes, respectively, where the second term in the ciphertext size accounts for the masked string u . Cycle counts and wall-clock times are measured on a single core of an Intel i7-1360P. The code is compiled with GCC 15.2, with Ofast, march=native and suggestions for vectorizations (OpenMP SIMD).

Benchmarks are run by repeating 10,000 times the API call; the code is run on a performance core. The arithmetic part of the code is competitive. A multiplication modulo q runs in $7 \mu\text{s}$ (respectively 14 and 31 for the larger proposals). A ring inversion runs in $45 \mu\text{s}$ (respectively 189, 198). The inversion modulo 2 is computed using a Newton iteration, and takes the time of roughly 3 full multiplications. The bias and the NTT of f are not stored in the secret key.

What dominates the running time of encapsulation/decapsulation is therefore hashing and random sampling.

Table 4: Performance of the reference implementation of YUANYANG.KEM on an Intel i7-1360P.

		YUANYANG.KEM-512	YUANYANG.KEM-1024	YUANYANG.KEM-2048
KeyGen	kcycles	439	1263	1725
	time (μs)	116	332	455
Encapsulate	kcycles	245	406	770
	time (μs)	65	107	203
Decapsulate	kcycles	293	529	1013
	time (μs)	77	139	267
pk size	Bytes	736	1568	3328
CT size	Bytes	656	1344	2880

7 Conclusion

This document specifies the NTRU-based KEM scheme – YUANYANG.KEM. The most distinctive feature of YUANYANG.KEM is the shared-key compatibility with its signature companion YUANYANG.DSA.

Shared-key compatibility with YUANYANG.DSA. YUANYANG.KEM uses the same parameters with YUANYANG.DSA, and the key pair of YUANYANG.DSA can be securely reused by YUANYANG.KEM with joint security. This property facilitates the applications in payment cards, secure group communication and constrained environments.

As a standalone general-purpose KEM proposal, YUANYANG.KEM has the following features in terms of security, efficiency and portability.

Conservative security. YUANYANG.KEM admits a formal theoretical security proof based on well-studied assumptions. Furthermore, YUANYANG.KEM adopts conservative parameters: its three parameter sets achieve the target security level under the Core-SVP model; its dense secret and randomness is beneficial to defeat the small-key attacks; the low decryption failure avoids the risk of failure-based attacks.

Good overall performance. YUANYANG.KEM has a good overall performance in terms of size and speed. Its size is smaller than the NIST post-quantum KEM standards Kyber and HQC at the same security level. Its key generation, encapsulation and decapsulation run fast. This makes YUANYANG.KEM suitable for most applications.

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